

JWST Galaxy Classification

Emission Line Analysis — Cambridge Research Summary

Overview

Stars in space give off radiation into the space around them, causing the surrounding gases and atoms to become ionized. By analyzing this radiation we can determine key features of that star or galaxy — including its age, chemical composition, and mass.

Key Variables: Metallicity and Mass

Two variables are central to understanding galaxy emission behaviour: metallicity and mass.

Mass: A larger stellar mass creates a stronger gravitational field, resulting in greater pressure at the star's core. This causes the star to consume its fuel faster, shortening its lifespan. Younger, more massive stars are more ionizing because they emit more radiation — producing more visible emission lines.

Metallicity: As the metallicity of a cloud increases, more elements and atoms are ionized, resulting in a greater number of emission lines observable in the spectrum.

The plot below shows how flux (Lux) varies with wavelength for galaxies at different metallicities. As a galaxy is younger and more metal-poor, its flux is higher.

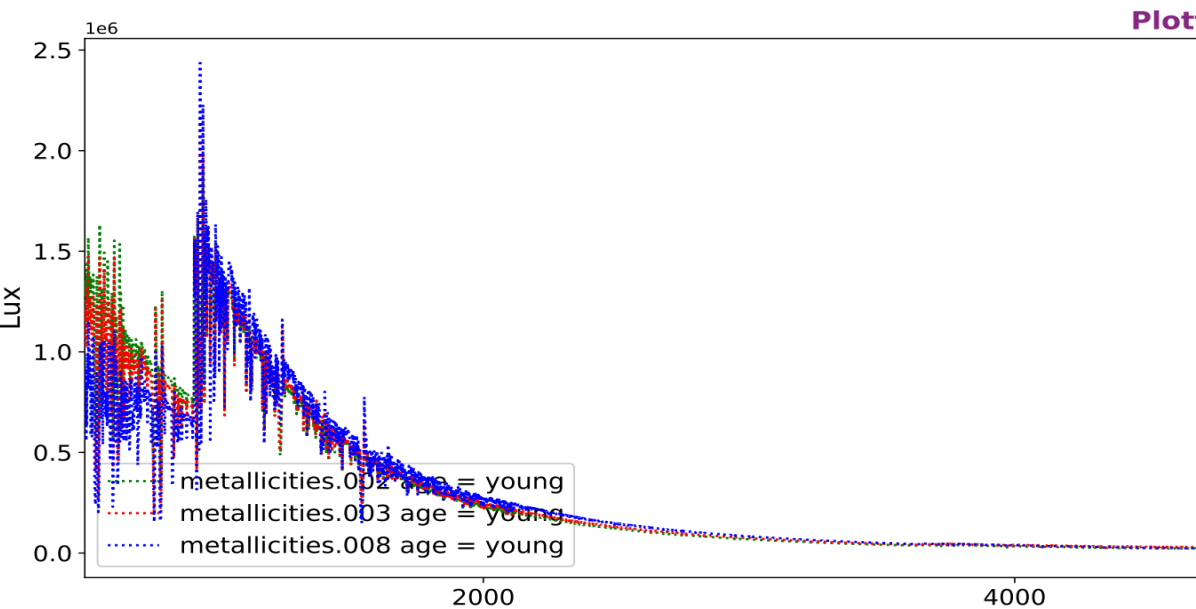


Figure 1: Flux vs wavelength for three metallicity values (0.002, 0.003, 0.008) at young stellar age. Higher flux corresponds to younger, more ionizing stars.

Emission Line Physics

Electromagnetic waves from a star can be split into spectra, making it possible to analyse both absorption and emission lines. When atoms gain energy, electrons jump to a higher energy level. As they fall back, they release an electromagnetic wave at a precise wavelength specific to that energy gap — giving each element a unique spectral fingerprint. This includes the Lyman-alpha ($\text{Ly}\alpha$) line, produced by hydrogen.

The CLOUDY Model

CLOUDY is simulation software that models galaxy radiation travelling through a cloud. By comparing simulated output to real observational data, we can estimate properties of the galaxy and its central star. The model outputs a spectrum which is overlaid on photometric data points to assess how closely the model matches reality.

The spectrum below shows the CLOUDY model output (green dotted line) overlaid with real photometric data (circles = spectroscopy, squares = photometry). Key emission lines — $\text{Ly}\alpha$, Oxygen, and $\text{H}\alpha$ — are marked with blue vertical lines.

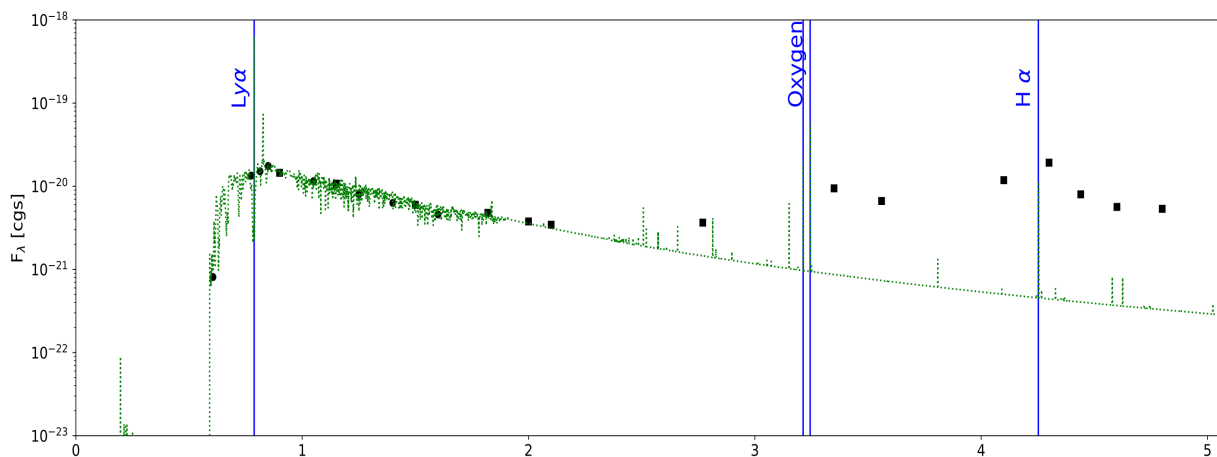


Figure 2: Labelled galaxy spectrum. Green dotted line = CLOUDY model. Black circles and squares = photometric data. Blue lines mark $\text{Ly}\alpha$, Oxygen, and $\text{H}\alpha$.

Challenges and Limitations

Redshift correction: JWST observes extremely distant objects, requiring wavelength adjustment for cosmological redshift. Corrected by multiplying CLOUDY data by $(1 + \text{redshift value})$.

Dust extinction: Dust and particles absorb energy and re-emit it as thermal radiation, shifting the observed spectrum redward and complicating analysis.

Limited photometric data: These galaxies are very distant and instruments are in high demand, resulting in as few as 5 data points per spectrum — making firm conclusions difficult.

CLOUDY model assumptions: CLOUDY assumes uniform density and a spherical hydrogen distribution — conditions that rarely hold in reality, particularly following a supernova. This limits the accuracy of model-to-data comparisons.